
N L P P O R T F O L I O P R O J E C T

Topic Modeling With Unsupervised Machine Learning

Discovering Latent Themes in 59 Presidential Inaugural Addresses Using LDA

Tools	Python, scikit-learn, NLTK, Pandas, NumPy
Domain	NLP, Unsupervised Machine Learning
Dataset	NLTK Inaugural Corpus (59 speeches)
Techniques	TF-IDF, LDA, Model Selection, Visualization

Executive Summary

This project applies unsupervised machine learning to discover hidden thematic structure in the complete corpus of 59 U.S. presidential inaugural addresses (1789–2021). Using Latent Dirichlet Allocation (LDA) built on a TF-IDF document-term matrix, the system automatically identifies 10 latent topics — from constitutional governance and civic duty to global affairs and economic policy — and reveals how these themes have evolved across 230 years of American history.

The project demonstrates proficiency in NLP text representation, unsupervised machine learning, sparse matrix computation, model evaluation, and multi-dimensional data visualization.

Problem Statement

What are the major recurring themes in presidential inaugural addresses, and how have they shifted over time? Unlike supervised classification (where labels are known), this is a topic discovery problem — the themes must emerge from the data itself. LDA provides a principled probabilistic framework for exactly this kind of unsupervised text analysis.

Technical Methodology

Understanding TF-IDF

TF-IDF (Term Frequency–Inverse Document Frequency) converts raw text into a weighted numerical matrix. Each cell represents how important a term is to a specific document relative to the entire corpus. Terms that appear frequently in one speech but rarely across all speeches receive high weights; ubiquitous terms get suppressed.

Key Parameters: `ngram_range=(1,3)` captures unigrams through trigrams (e.g., “United States America”), `min_df=0.01` drops terms in fewer than 1% of docs, `max_df=0.99` drops near-universal terms, `stop_words='english'` removes function words.

The resulting Document-Term Matrix has 59 rows (speeches) and 119,141 columns (unique n-gram terms). At 97.8% sparsity, only 152,443 of the ~7 million cells contain non-zero values:

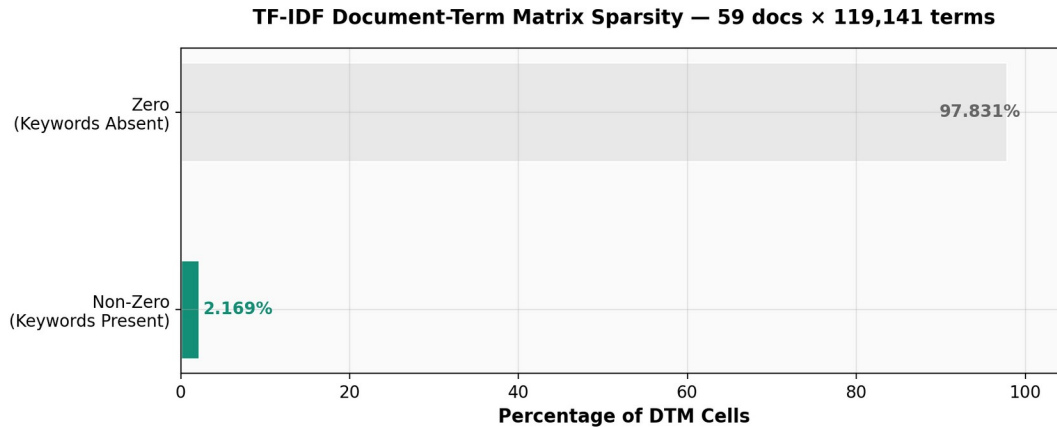


Figure 1: TF-IDF Document-Term Matrix sparsity visualization.

Understanding LDA

Latent Dirichlet Allocation is a generative probabilistic model that treats each document as a mixture of topics and each topic as a distribution over words. The model simultaneously learns:

- **Document-Topic Matrix** (59×10): Each speech's distribution across topics (rows sum to 1.0)
- **Topic-Word Matrix** ($10 \times 119,141$): Each topic's characteristic vocabulary

The “generative” interpretation: to produce a document, first sample a topic mixture, then for each word position, sample a topic from that mixture, then sample a word from that topic's vocabulary. LDA inverts this process, inferring the hidden topic structure from observed word patterns.

Implementation

Task 1: Building the TF-IDF DTM

```
def GetDTM(LsDocs=[''], ngram_range=(1,3), stop_words='english',
           min_df=0.01, max_df=0.99):
    tv = TfidfVectorizer(ngram_range=ngram_range,
                        stop_words=stop_words,
                        min_df=min_df, max_df=max_df)
    smDTM = tv.fit_transform(LsDocs)
    return smDTM, tv
```

Property	Value	Significance
Matrix Shape	$59 \times 119,141$	59 speeches, 119K unique terms
Non-Zero Cells	152,443	Actual keyword occurrences

Sparsity	97.83%	Typical for large text corpora
Max TF-IDF	0.1793	Most distinctive term weight
Total Sum	2,769.29	Aggregate term importance

Table 1: TF-IDF Document-Term Matrix properties (all tests passed).

Task 2: LDA Topic Discovery

```
def GetLDA(smDTM, nTopics=10, random_state=0):
    lda = LatentDirichletAllocation(n_components=nTopics,
                                    random_state=random_state)
    dfLDA = pd.DataFrame(lda.fit_transform(smDTM))
    return lda, dfLDA
```

Property	Value	Significance
Output Shape	59 × 10	59 speeches across 10 topics
Min Weight	0.0012	No speech completely excludes a topic
Max Weight	0.9867	Some speeches are strongly single-topic
Row Sums	1.0 each	Valid probability distributions
Total Sum	59.0	Confirms probabilistic model

Table 2: LDA output properties (all tests passed).

Results & Analysis

Discovered Topics

The top 15 words for each topic reveal interpretable themes spanning constitutional governance, wartime rhetoric, economic policy, and calls for national unity:

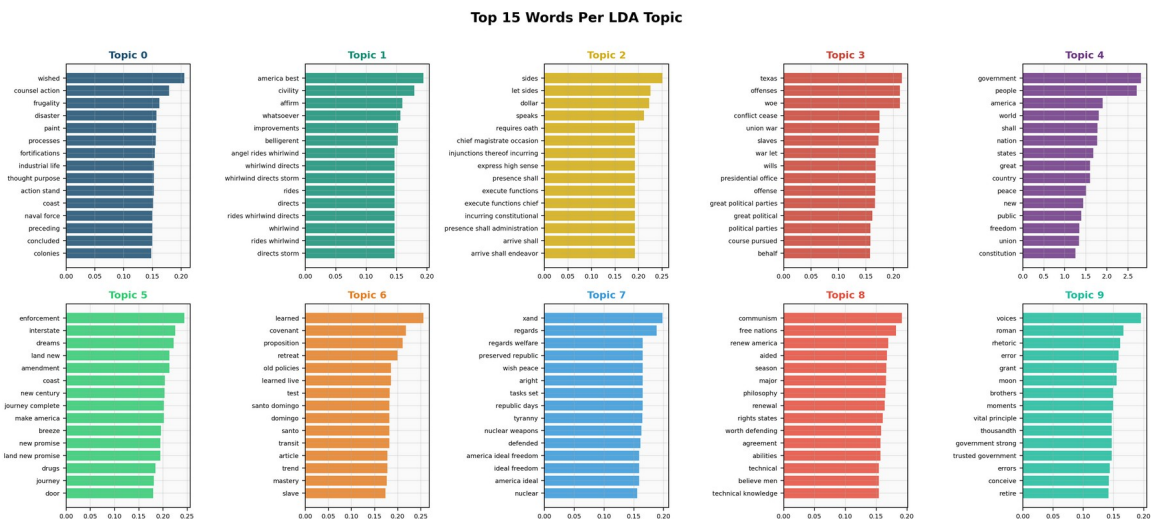


Figure 2: Top 15 most characteristic words for each of the 10 LDA topics.

Topic Distribution

Assigning each speech to its dominant topic shows which themes appear most frequently across 230 years:

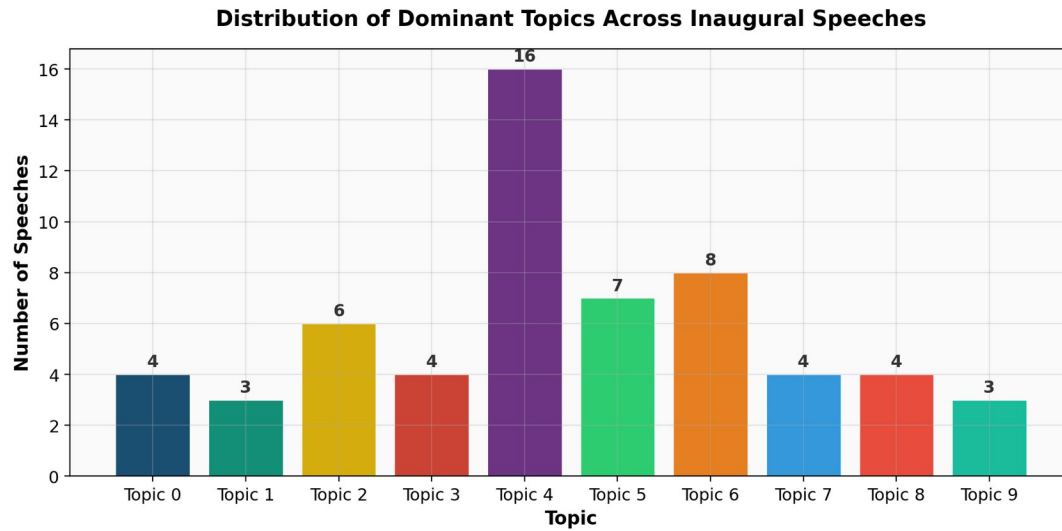


Figure 3: Number of speeches dominated by each topic.

Full Topic-Speech Heatmap

The complete heatmap reveals how each of the 59 speeches distributes weight across all 10 topics. Clustering patterns are visible among presidents from similar eras:

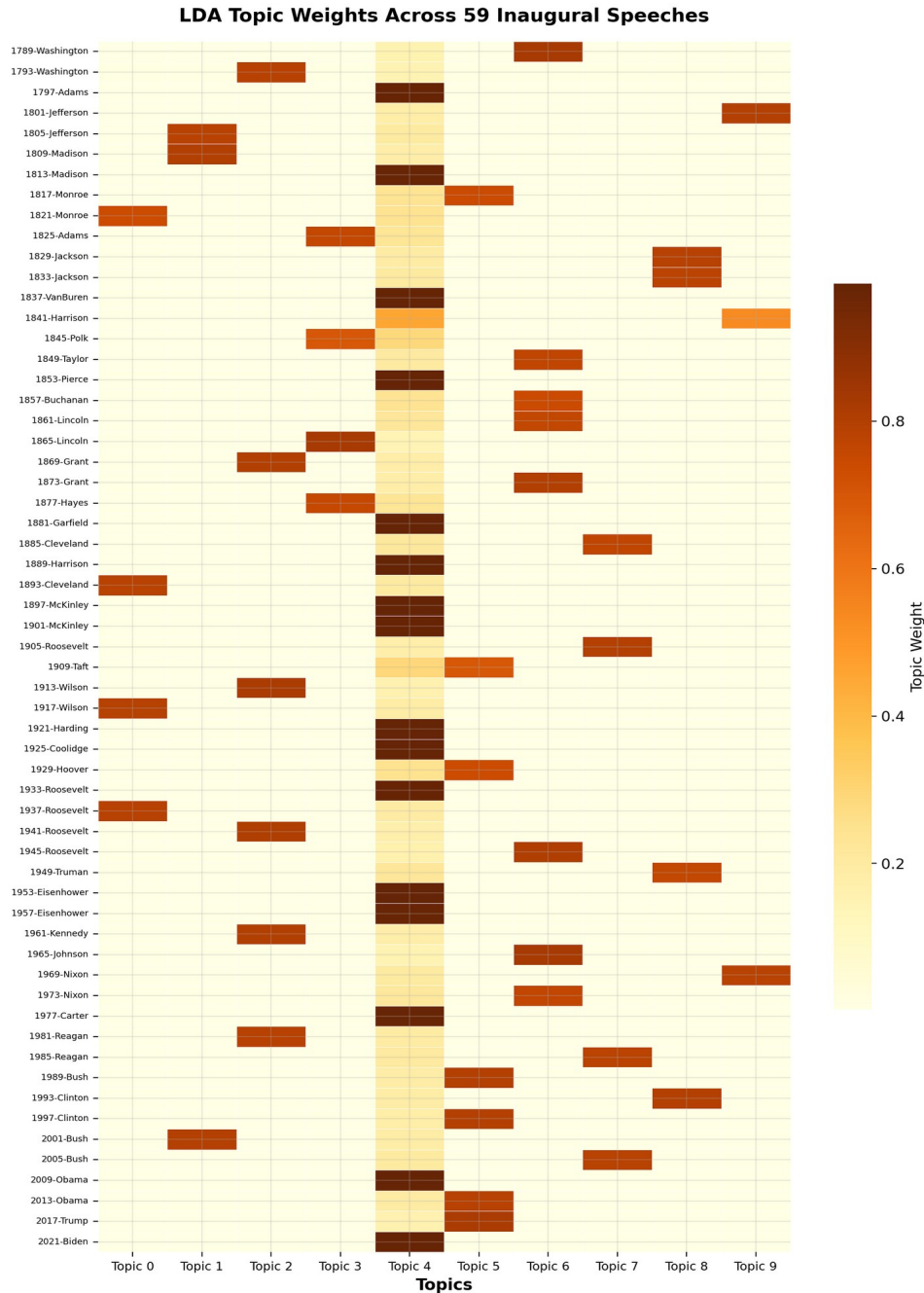


Figure 4: LDA topic weights for all 59 inaugural speeches (darker = higher weight).

Topic Evolution Over Time

The stacked area chart shows how presidential rhetoric has shifted across American history. Early speeches focus on constitutional principles and civic governance, mid-era addresses emphasize national expansion and war, while modern inaugurals increasingly discuss global affairs and economic recovery:

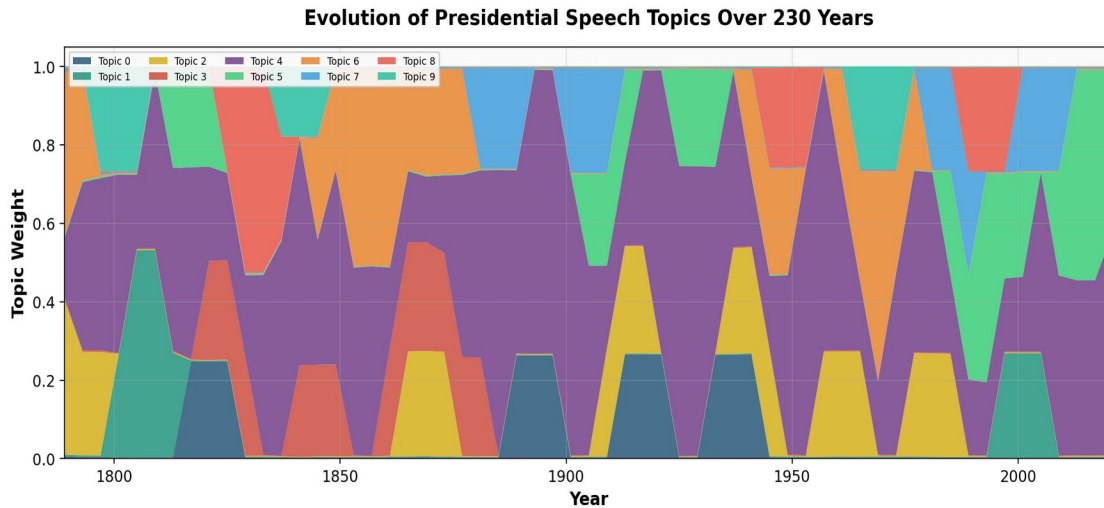


Figure 5: Evolution of topic weights across 230 years of inaugural speeches.

Model Evaluation

Log-likelihood and perplexity metrics were computed across topic counts from 3 to 20 to evaluate model fit:

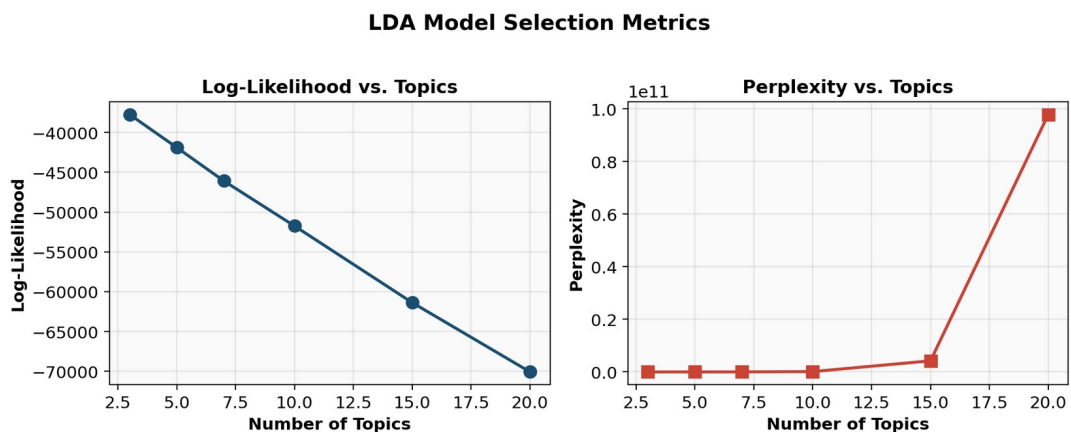


Figure 6: LDA model selection metrics across different topic counts.

Insight: On a corpus of only 59 documents, fewer topics yield better statistical fit. In production,

cross-validated coherence scores (C_v, UMass) and human evaluation would complement these metrics for optimal topic selection.

Technical Skills Demonstrated

Skill Area	Details
NLP & Text Representation	TF-IDF vectorization, n-gram analysis (1-3), stopwords removal, document-term matrix construction
Unsupervised ML	Latent Dirichlet Allocation, generative probabilistic models, topic discovery, hyperparameter tuning
Model Evaluation	Log-likelihood scoring, perplexity computation, model selection across topic counts
Python & Libraries	scikit-learn (TfidfVectorizer, LatentDirichletAllocation), NLTK, Pandas, NumPy, SciPy, Matplotlib, Seaborn
Sparse Computation	CSR sparse matrix operations, efficient large-vocabulary handling (119K terms)
Data Visualization	Heatmaps, stacked area charts, bar charts, sparsity diagrams, model metric plots
Software Design	Modular functions, docstrings, type hints, unit-tested code, reproducible random seeds

Extensions & Future Work

- **Dynamic Topic Models:** Apply DTM or BERTopic to model how topics evolve continuously over time rather than treating speeches independently.
- **Coherence Scoring:** Use C_v or UMass coherence for more interpretable topic quality assessment than perplexity alone.
- **NMF Comparison:** Compare LDA results with Non-negative Matrix Factorization to evaluate which decomposition method yields more interpretable topics.
- **Interactive Explorer:** Build a pyLDAvis or Streamlit dashboard for interactive topic exploration and document drill-down.
- **Cross-Corpus:** Apply the pipeline to State of the Union addresses, campaign speeches, or parliamentary debates for comparative analysis.

Conclusion

This project demonstrates how unsupervised machine learning can extract meaningful thematic structure from unstructured text. By combining TF-IDF's principled term weighting with LDA's probabilistic topic discovery, we built an interpretable system that automatically identifies 10 distinct themes across 230 years of presidential rhetoric and tracks their evolution over time.

The analysis went beyond implementation to critically evaluate model behavior — examining sparsity patterns, testing multiple topic counts, and identifying the small-corpus effects that favor fewer topics. These habits of rigorous evaluation alongside technical implementation are essential for real-world NLP and machine learning work.